

Hao Xu

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Research Interests

Foundation models for decision-making • Partially Observable Markov Decision Processes (POMDPs) • LLM applications in manufacturing and materials science

Education

Ph.D. in Mathematics and Statistics

2025–Present

The University of Sydney, Sydney, Australia

Research: Foundation models for decision-making, POMDPs, and AI/LLM applications in manufacturing and materials science.

MPhil in Smart Manufacturing

2022–2024

HKUST (Guangzhou), Guangzhou, China

Research: Anomaly detection in industrial manufacturing; high-dimensional data processing for point clouds and images.

M.S. in Data Science

2021–2022

City University of Hong Kong, Hong Kong, China

Research: Deep learning and predictive modeling; attention-based financial sentiment analysis and epidemiological mortality monitoring.

B.Eng. in Materials Science and Engineering

2016–2020

Chongqing University, Chongqing, China

Research: Materials synthesis, characterization, and failure analysis; ODS steels, Mg-based hydrogen storage alloys, and high-voltage testing. **Grade Rank: Top 5%.**

Publications

* indicates co-first authors; † indicates corresponding authors.

- [1] Xiangru Jian*, **Hao Xu***†, Wei Pang*, Xinjian Zhao, Chengyu Tao, Qixin Zhang, Xikun Zhang†, Chao Zhang, Guanzhi Deng, Alex Xue, Juan Du, Tianshu Yu, Garth Tarr, Linqi Song, Qiuzhuang Sun, Dacheng Tao.
FORGE: Fine-grained Multimodal Evaluation for Manufacturing Scenarios.
Findings of the Association for Computational Linguistics: EMNLP 2026. [\[arXiv\]](#)
- [2] **Hao Xu***, Xiangru Jian*†, Xinjian Zhao*, Wei Pang*, Chao Zhang, Suyuchen Wang, Qixin Zhang, Zhengyuan Dong, Joao Monteiro, Bang Liu, Qiuzhuang Sun, Tianshu Yu†.
GraphOmni: A Comprehensive and Extendable Benchmark Framework for Large Language Models on Graph-theoretic Tasks.
International Conference on Learning Representations (ICLR), 2026. [\[arXiv\]](#)
- [3] Chengyu Tao, **Hao Xu**, Juan Du†.
F2PAD: A General Optimization Framework for Feature-Level to Pixel-Level Anomaly Detection.
arXiv preprint arXiv:2407.06519, 2024. [\[arXiv\]](#)
- [4] **Hao Xu**, Juan Du†, A. Wang, Y.C. Chen.
Ano-SuPs: Multi-size Anomaly Detection for Manufactured Products by Identifying Suspected Patches.
Accepted in *INFORMS Journal on Data Science*, 2023. [\[arXiv\]](#)

Selected Awards

Finalist, Best Paper Competition	INFORMS Conference on Quality, Statistics, and Reliability	2025
Outstanding Graduate of Chongqing City		2020
National College Students Summer Social Practice Special Action Bronze Award (<i>Deputy Leader</i>)		2018